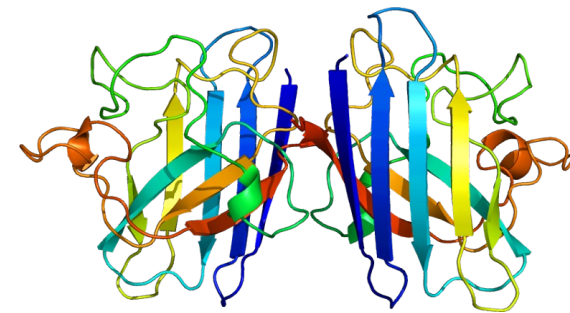
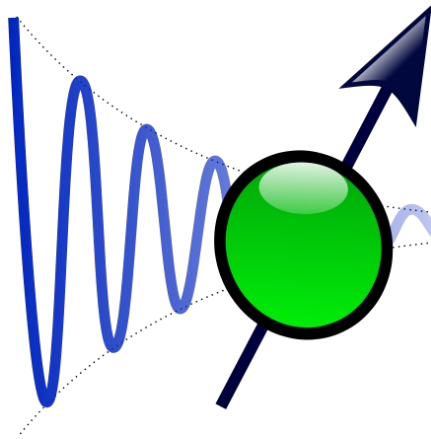


CHEMICAL SCIENCES



SPECTRUM BETWEEN PHYSICS ON ONE HAND AND BIOLOGY ON THE OTHER.

Physical chemistry deals with the principles of chemistry/physics involved in chemical interactions.

It examines:

- How chemical reactions occur ? (Thermodynamics)
- How much of product is formed ? (Mole concepts/ Equilibrium)
- How much of product is formed at particular time? (Kinetics)
- How matter behaves on a molecular and atomic level (quantum)

Law of Conservation of Mass (1778)

In 1778 Lavoisier found that when mercury oxide is heated its weight decreases. The oxygen gas it releases has exactly the same weight as the weight lost by the mercury oxide.



Law of Conservation of Mass (1778)

- Mass is neither created nor destroyed in chemical reactions. In other words, the mass of any one element at the beginning of a reaction will equal the mass of that element at the end of the reaction.
- Lavoisier's finding laid the foundation for modern chemistry and revolutionized science.



Law of Conservation of Mass (1778)

Working again with Pierre-Simon Laplace, Lavoisier burned hydrogen with oxygen and found that water was produced, establishing that water is not an element, but is actually a compound made from the elements hydrogen and oxygen. This result astonished many people, because at that time 'everyone knew' that water was itself one of the 'indivisible' elements.



John Dalton

In 1808, [John Dalton](#) published his atomic theory proposing that all matter is made of atoms. He further stated that all atoms of an element are identical, and the atoms of different elements have different masses.

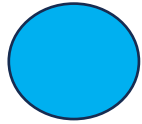


John Dalton

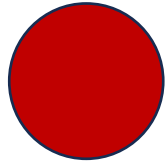
Chemical combination occurs by the union of the atoms of the elements in simple numerical ratios.



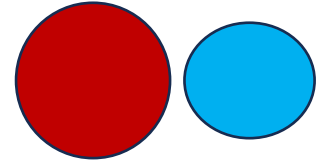
A simple Reaction



Hydrogen



Oxygen



John Dalton

- The first scientists to determine relative atomic masses were [John Dalton](#) and [Thomas Thomson](#) between 1803 and 1805 and [Jöns Jakob Berzelius](#) between 1808 and 1826.

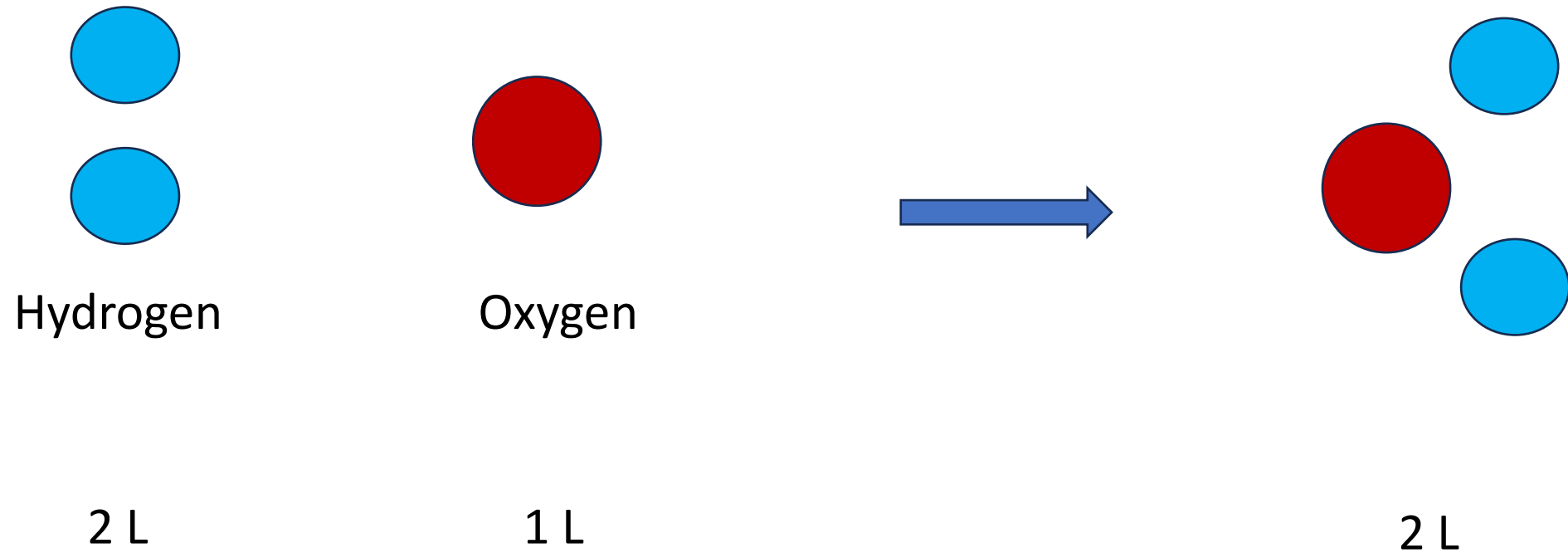


Joseph Gay-Lussac

- In 1809 Joseph Gay-Lussac published his law of combining gas volumes. He had noticed that when two liters of hydrogen gas react with one liter of oxygen gas, they form two liters of gaseous water at same temperature and pressure. All gases that he reacted seemed to react in simple volume ratios.



A simple Reaction





Avogadro (1811)

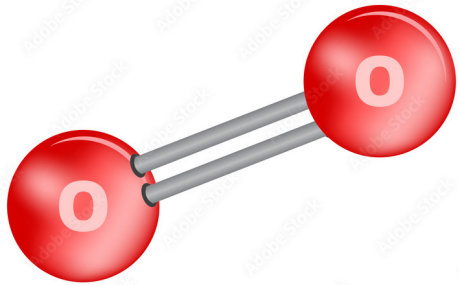
He said that the best explanation for Gay-Lussac's observations of gas reactions was that equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules. This is now called Avogadro's law.



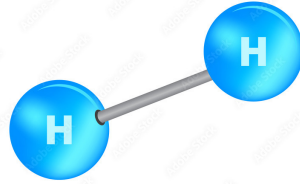
Avogadro (1811)

The reason that two liters of hydrogen gas react with a liter of oxygen gas to form just two liters of gaseous water is that the volume decreases because the number of particles present decreases.

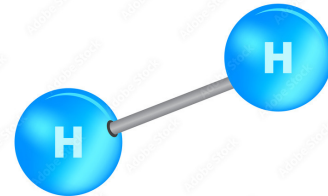
A simple Reaction



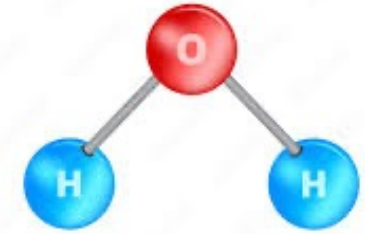
Oxygen | O_2



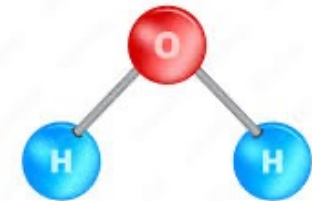
Hydrogen | H_2



Hydrogen | H_2



Water | H_2O



Water | H_2O

1 L of oxygen will react with 2 L of hydrogen to produce 2 L of gaseous water



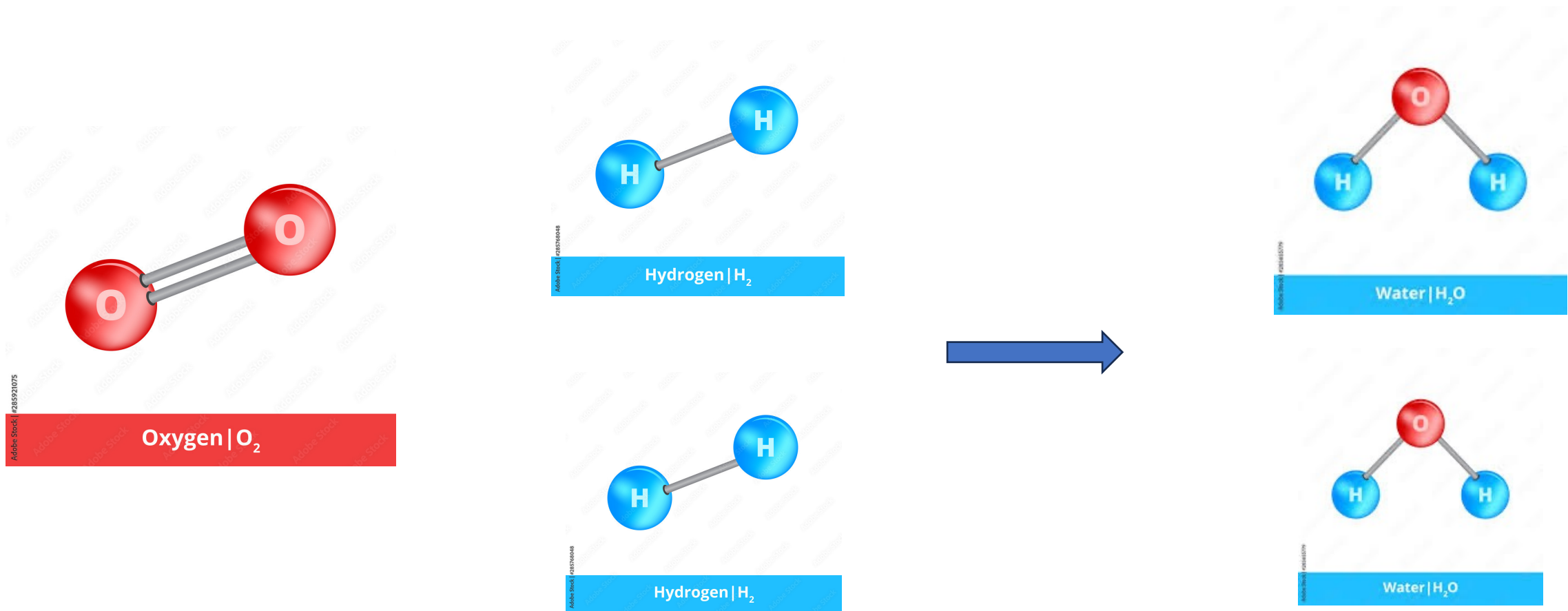
Avogadro (1811)

As a result of these observations Avogadro became the first scientist to realize that elements could exist as molecules rather than as individual atoms. For example, he recognized that the oxygen around us exists as a molecule in which two atoms of oxygen are linked.



Wilhelm Ostwald (1900)

- Ostwald defined the mole as 'that amount of any gas that occupies a volume of 22,414 mL in normal conditions is called one mole'



1 L of oxygen will react with 2 L of hydrogen to produce 2 L of gaseous water

22.4 L of oxygen will react with 44.8 L of hydrogen to produce 44.8 L of gaseous water

1 mole of oxygen will react with 2 mole of hydrogen to produce 2 mole of gaseous water

Can you count it ??



Jean Perrin

- The name *Avogadro's number* was coined in 1909 by the physicist [Jean Perrin](#), who defined it as the number of molecules in exactly 32 grams of [oxygen](#) gas.^[14]
- The goal of this definition was to make the mass of a mole of a substance, in grams, be numerically equal to the mass of one molecule relative to the mass of the hydrogen atom; which, because of the [law of definite proportions](#), was the natural unit of atomic mass, and was assumed to be 1/16 of the atomic mass of oxygen.



Definition: Avogadro's Number

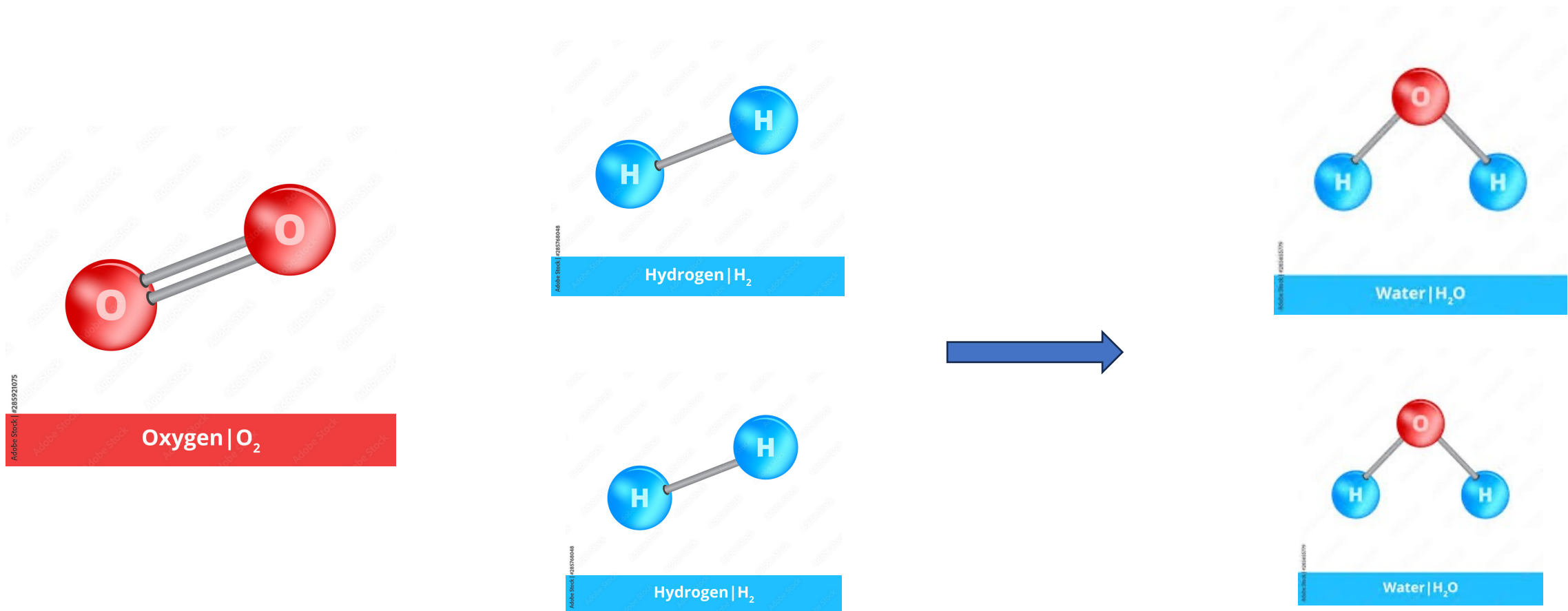
Avogadro's number is equal to $6.02214076 \times 10^{23}$.

Definition: Mole (mol)

The mole is the SI unit of the amount of a substance:

1 mole = $6.02214076 \times 10^{23}$ entities.

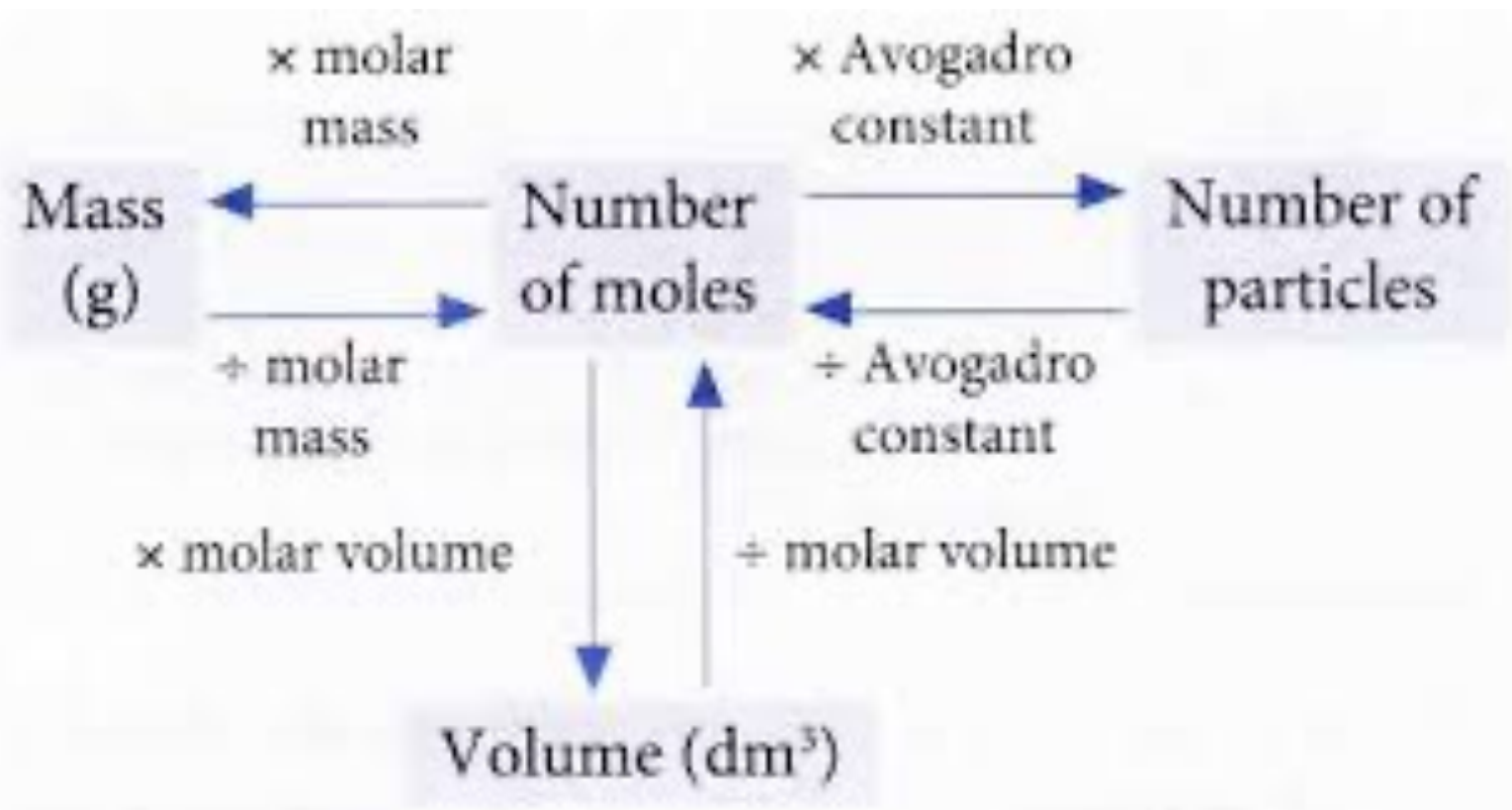
As a unit, the mole functions in a similar way to the unit dozen. We know that one dozen is equal to 12 items. If there are 12 bananas on a table, we could say that there is one dozen bananas. Similarly, if there are $6.02214076 \times 10^{23}$ molecules of oxygen, we could say that there is one mole of oxygen.



22.4 L of oxygen gas will react with 44.8 L of hydrogen gas to produce 44.8 L of gaseous water

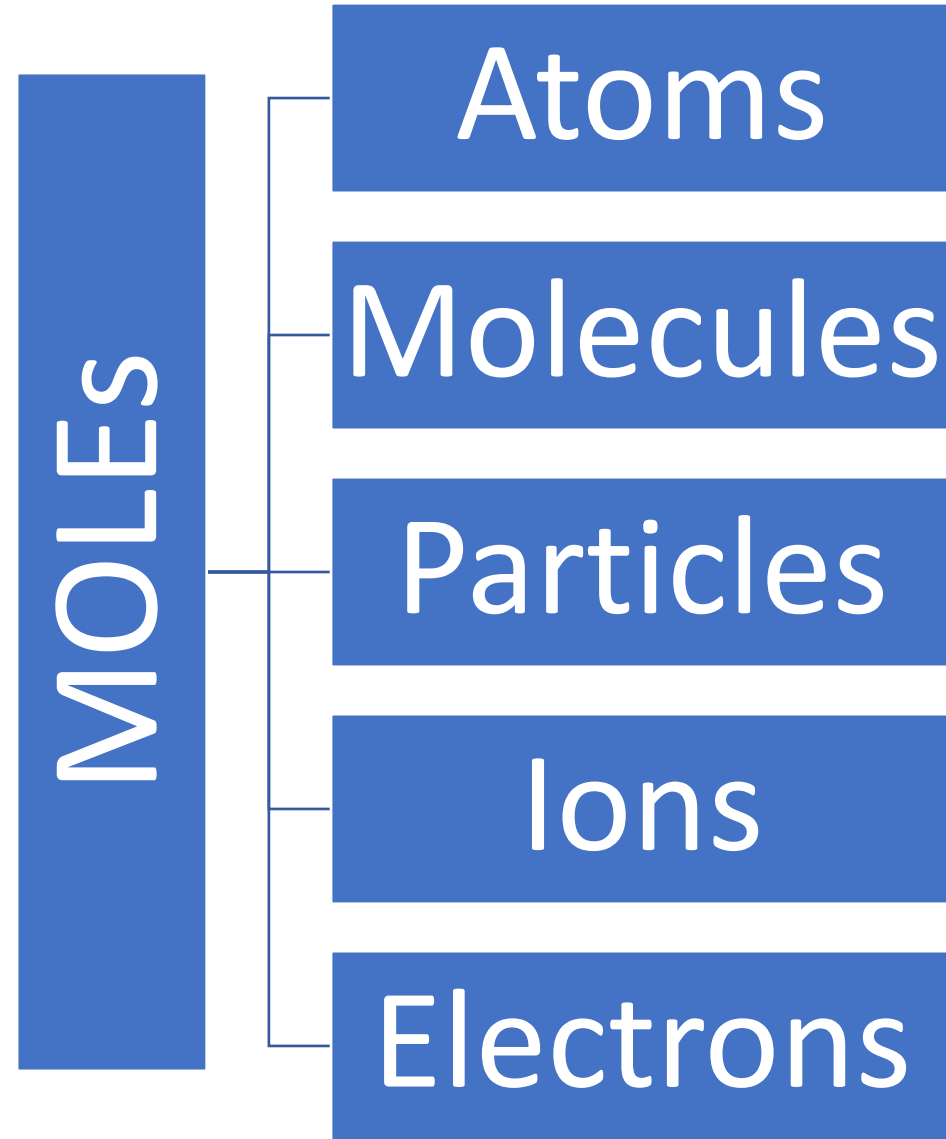
1 mol of oxygen gas will react with 2 mol of hydrogen gas to produce 2 mole of gaseous water

32 g of oxygen gas will react with 4 g of hydrogen gas to produce 36 g of gaseous water



Refinement in the definition of Atomic weight

- In the 1860s, [Stanislao Cannizzaro](#) refined relative atomic masses by applying [Avogadro's law](#) (notably at the [Karlsruhe Congress](#) of 1860). He formulated a law to determine relative atomic masses of elements: *the different quantities of the same element contained in different molecules are all whole multiples of the atomic weight* and determined relative atomic masses and molecular masses by comparing the [vapor density](#) of a collection of gases with molecules containing one or more of the chemical element in question.



Can you count it ??

